

Q Given an array.
Find the no. of Special indexes!

→ after deleting that element

$$\boxed{\text{Sum of all even indexed elements} = \text{Sum of all odd indexed elements}}$$

Ex
A: ⁰4 ¹3 ²2 ³7 ⁴6 ⁵-2

del(0)

A: ⁰3 ¹2 ²7 ³6 ⁴-2

$$S_o = 8$$

$$S_e = 8$$

del(1)

A: ⁰4 ¹2 ²7 ³6 ⁴-2

$$S_o = 8$$

$$S_e = 9$$

del(2)

A: ⁰4 ¹3 ²7 ³6 ⁴-2

$$S_o = 9$$

$$S_e = 9$$

3, 4, 5
- - -



Se

So

Array!

Idea:

$$S_e = S_e[0, i-1] + S_o[i+1, N-1]$$

$$\begin{matrix} PS[i] - PS[i-1] \\ PS[N-1] - PS[i+1-1] \end{matrix}$$

$$S_e = PS_e[i-1] + PS_o[N-1] - PS_o[i]$$

$$S_o = S_o[0, i-1] + S_e[i+1, N-1]$$

$$S_o = PS_o[i-1] + PS_e[N-1] - PS_e[i]$$

STEPS

1. Calc. the PS_{old} and PS_{even} Array. $\rightarrow O(N)$

2. $cnt = 0;$

$\{ i = 0; i < N; i++ \}$

$$S_e = PS_e[i-1] + PS_o[N-1] - PS_o[i];$$

$$S_o = PS_o[i-1] + PS_e[N-1] - PS_e[i];$$

$\text{if } (S_o == S_e) \{$
 $\quad cnt++;$

$\}$

$\}$
 $\text{return } cnt;$

$\boxed{TC = O(N)}$
 $\boxed{SC = O(N)}$

$$A: \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 \\ 4 & 3 & 2 & 7 & 6 & -2 \end{matrix} \quad N=6$$

$$PSc = \begin{matrix} 4 & 4 & 6 & 6 & 12 & 12 \end{matrix}$$

$$PS_0 = \begin{matrix} 0 & 3 & 3 & 10 & 10 & 8 \end{matrix}$$

$$A: \begin{matrix} 0 & 1 & 2 & 3 & 4 \\ 4 & 2 & 7 & 6 & -2 \end{matrix}$$

del (1)

$$S_0 = PS_0[i-1] + PSe[N-1] - PSe[i] \\ = PS_0[0] + PSe[5] - PSe[1] = 0 + 12 - 4 = 8$$

$$S_c: PS_c[i-1] + PS_o[N-1] - PS_o[i]$$
$$= PS_c[0] + PS_o[5] - PS_o[1] = 4 + 8 - 3 = 9$$

Another idea
2 variables

A:

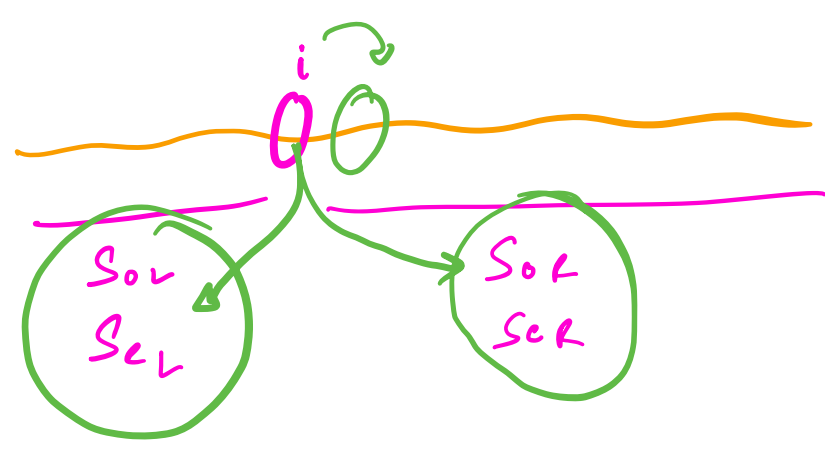
0	1	2	3	4	5
3	5	7	6	7	8

$= 15 + 6$
 L

$= 21 - 6$
 R

N

A



HW →

Do it in $O(1)$ space!

Q Given an array of N elements.
Find the majority element!

↳ An element with $\text{freq} > N/2$

$A: [1, 2, 1, 6, 1, 1]$

$N=6$

$$N/2 = 6/2 = 3$$

ME: 1, $\text{freq}(1) = 4$

$$4 > 3$$

$A: [1, 2, 2, 6, 1, 1]$

No ME! No element having $\text{freq} > 3$

$A: [3, 3, 4, 6, 1, 3, 2, 5, 3, 3, 3, 3]$

$N=12$

ME: 3, $\text{freq}(3) = 7 > 12/2$ ✓

$A: [1, 3, 4, 2, 1]$

I) BF

✗ elements find their freq
check if $> N/2$

$$\boxed{TC = O(N^2)}$$

$$\boxed{SC = O(1)}$$

II) Sorting

$A = [1, 3, 1, 5, 3, 3, 5]$

$Sort(A) : [1, 1, 3, 3, 3, 3, 5, 5]$

$TC = O(N \log N + N)$

$TC = O(N \log N)$

$SC = O(1)$

Heap Sort

III) Hash Map

$A : [1, 3, 1, 5, 3, 3, 5]$

NO ME

$TC : O(N)$

$SC : O(N)$

$\langle \text{key}, \text{value} \rangle$
 $\langle \text{element}, \text{freq} \rangle$

$\langle 1, 2 \rangle$
 $\langle 3, 3 \rangle$
 $\langle 5, 2 \rangle$

$N = 7$
 $7^{7/2}$
 $7^{3.5}$
 714

$7^{1/2}$

IV

MOORE'S VOTING ALGORITHM

Q Can there be more than 1 ME?

$x, y \rightarrow ME$

$x \neq y$

$$f(x) > N/2$$

$$f(y) > N/2$$

$$\hline f(x) + f(y) > N \quad X$$

→ AT MAX ONE M.E.!

~~A: [1 3 1 5 3 3 3]~~
 $\downarrow$
 $[1, 3, 5, 3, 3]$

N	<u>ME</u>	<u>$f_r(ME)$</u>
7	3	4
5	3	3

○ If I del 1 row of ME ✓
1 — NME ✓

→ the ME does not change!



$A: [1 \quad 3 \quad 3 \quad 1 \quad 3 \quad 5 \quad 3]$
 $\downarrow$
 $[1, 3, 3, 3, 3]$

→ If I del 2 Diff nrt's [1 occ.]
→ the nrt does not change!

→ If I del ANY 2 diff. elements
the ME. would NOT change!

A: [~~1~~ ~~7~~ ~~3~~ ~~8~~ ~~8~~ ~~4~~ ~~5~~ ~~7~~ ~~9~~]

ME: 4

A: [~~4~~ ~~3~~ ~~3~~ ~~2~~ ~~1~~ ~~4~~ ~~4~~ ~~5~~ 4 4 7]
ME: 4

A: [~~1~~ ~~2~~ ~~7~~ ~~1~~ ~~4~~ ~~5~~ ~~1~~ ~~4~~ ~~1~~ ~~4~~] $N = 10$
No ME!

[~~2~~ ~~2~~ ~~2~~ ~~2~~ ~~3~~ ~~3~~ ~~3~~ ~~3~~] No ME

A: [4 4 ~~3~~ ~~8~~ 8 ~~4~~ 9 ~~4~~ 4]
 CME: x 4 ↓ 4 → 4 → x → 8 → x → 9 → x → 4
 freq: 0 1 → 2 → 1 → 0 → 1 → 0 → 1 → 0 → 1

Validate:

$f(4) = 5 > 9/2$ ✓

4: ME

A: [1 ~~2~~ 4 ~~3~~ 3 3]

CME: $x \rightarrow 1 \rightarrow x \rightarrow 4 \rightarrow x \rightarrow 3$

freq: $0 \rightarrow 1 \rightarrow 0 \rightarrow 1 \rightarrow 0 \rightarrow 1 \rightarrow 2$

Validate: $f(3) = 3 \neq \frac{6}{2}$

NO. ME

A: [1 1 1 ~~3~~ 1 ~~4~~ ~~2~~ 1]

CME: $x \rightarrow 1 \rightarrow 1 \rightarrow 1 \rightarrow 1 \rightarrow 1 \rightarrow 1 \rightarrow 2$

freq: $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 2 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 2$

Validate: $f(1) = 5 \neq \frac{8}{2}$

1: ME ✓

```

cme = -1, f = 0;
for (i = 0 → N-1) {
    if (f == 0) {
        cme = A[i];
        f = 1;
    }
    else {
        if (cme == A[i]) {
            f++;
        }
        else {
            f--;
        }
    }
}

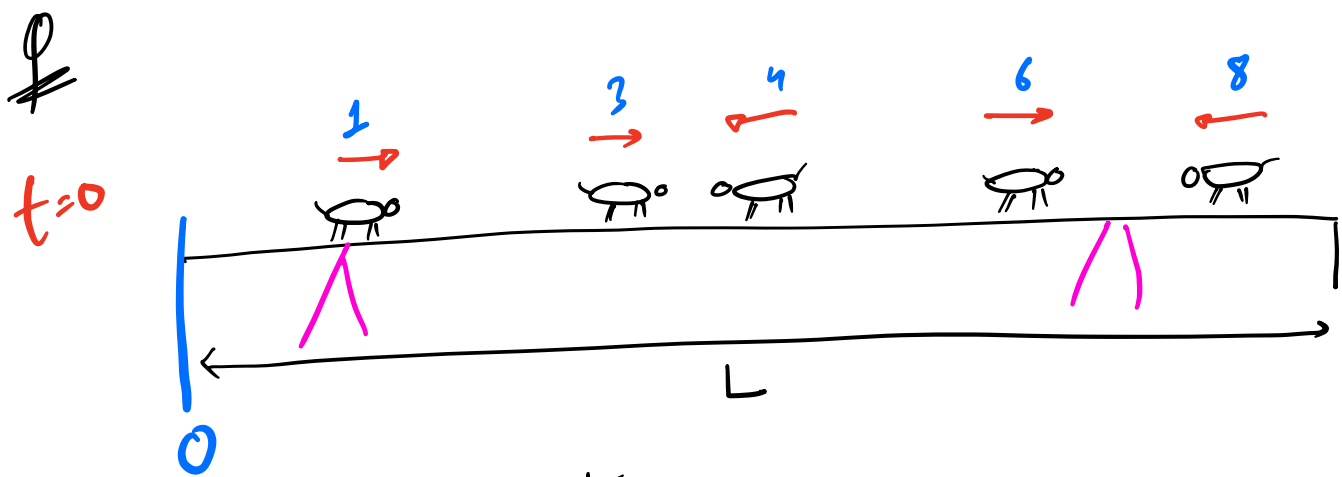
```

~~TC = $O(N)$~~
~~SC = $O(1)$~~

Validate if
 $f(\text{cme}) > N/2$

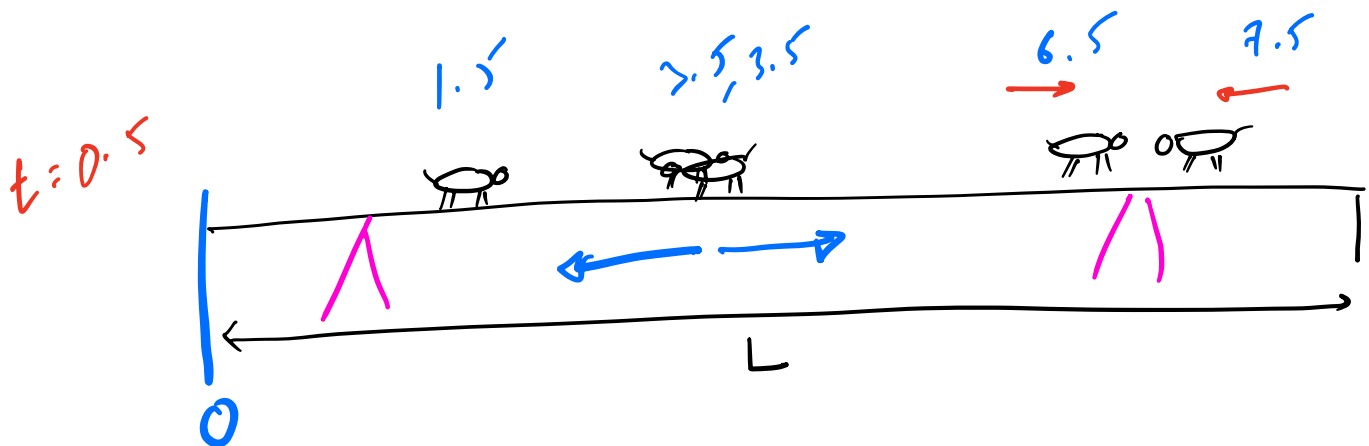
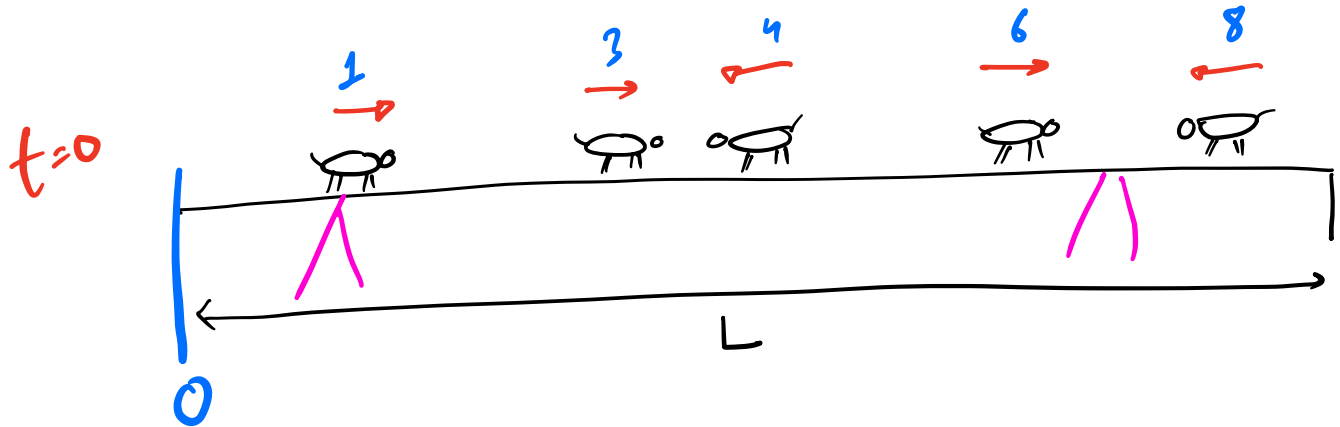
: true: return cme;

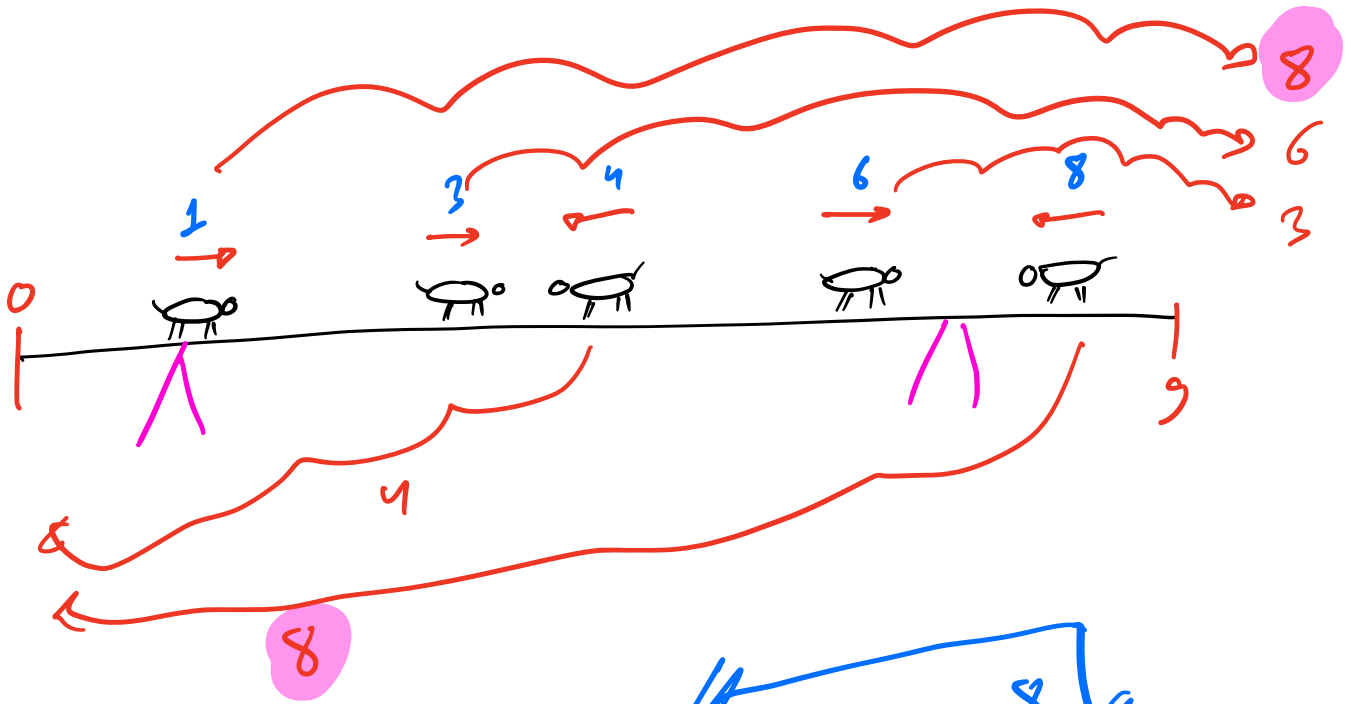
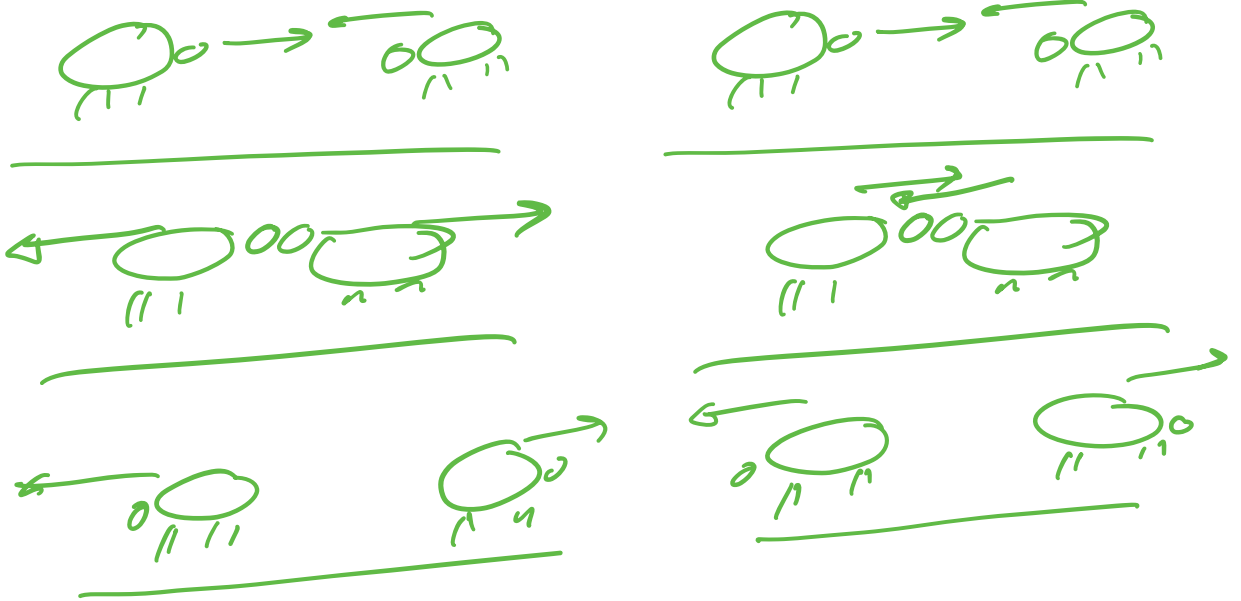
: false: return -1;



Q → After how much time will all the ANTS fall off?

Speed of ANTS $= 1 \text{ cm/sec}$





ANS = 8

Q Given an $A[]$
 — q queries $\rightarrow [L, R]$
 find the sum of even indexed elements in $[L, R]$

Q		
L	R	
0	3	$\rightarrow 4$
0	0	$\rightarrow 1$
2	4	$\rightarrow 10$

$A = [1, 3, 2, 2, 7]$
 $A' = [1, 0, 3, 0, 7]$
 $PS = [1, 1, 4, 4, 11]$

Idea: $A'[i] = 0$: i is odd $\rightarrow O(N)$
 $A'[i] = A[i]$: i is even

Create $PS[]$ for the $A'[]$ $\rightarrow O(N)$

Answer queries using $PS[]$ $\rightarrow O(Q)$

$$\boxed{\text{sum}(L, R) = \text{ps}[R] - \text{ps}[L-1]}$$

$$\downarrow$$

$$\boxed{L == 0} ?$$

$$\boxed{\text{TC} = O(N+Q)}$$

$$\boxed{\text{SC} = O(N)} \rightarrow O(1)$$

QUIZ

A: [⁰2, ¹~~4~~, ²3, ³~~1~~, ⁴5]

ps: [2, 2, 5, 5, 10] ✓

I Same Qus for odd-inclined elements

SIMILAR ANS AS ABOVE